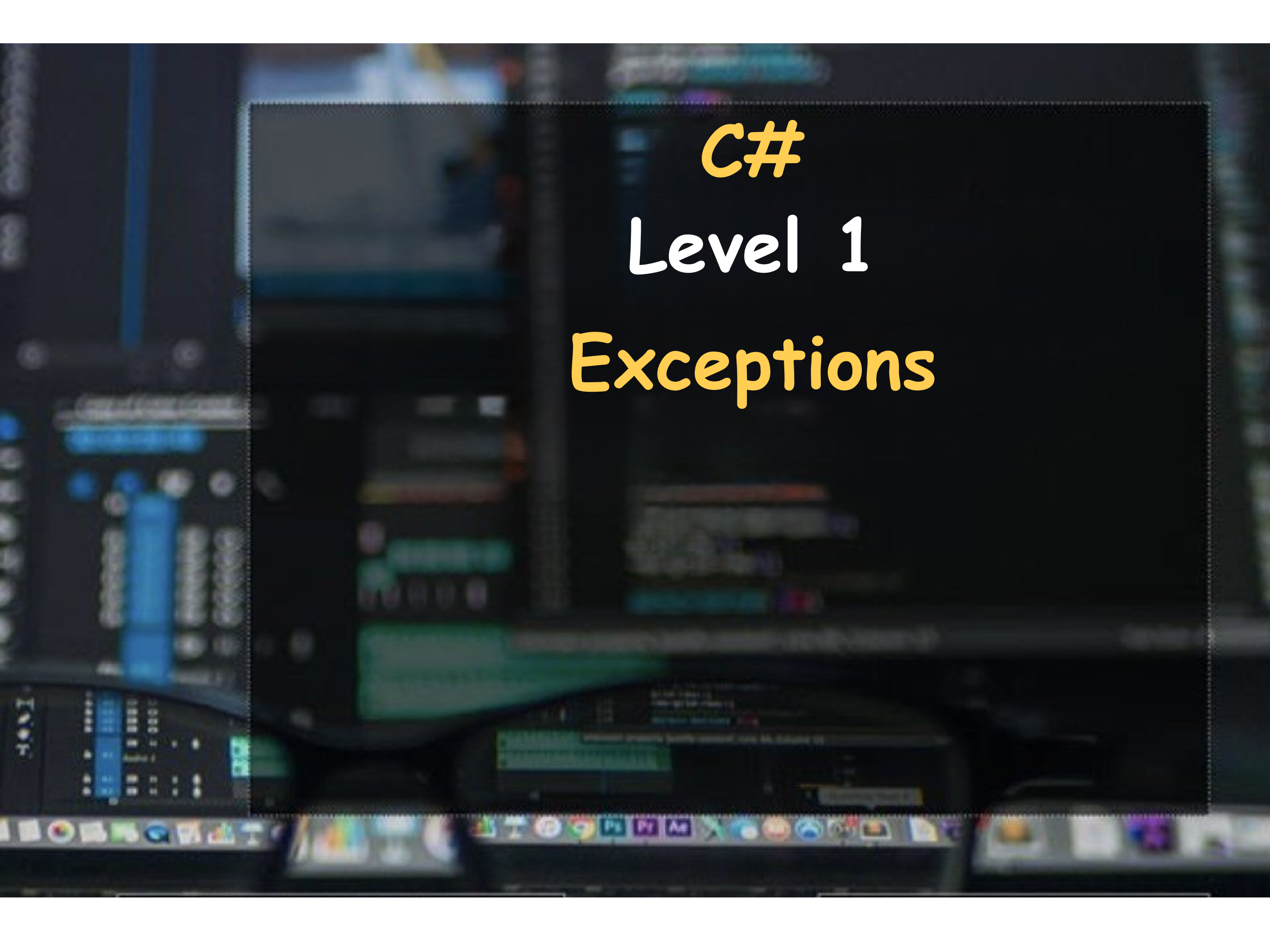




C#

Level 1

Exceptions



C#

Level 1

Exceptions

Definition

Exception

-  An exception is an error that occurs while the program is running.
-  C# throws an exception when an unexpected error occurs.
-  Without handling, an unhandled exception can terminate the program.

try...catch

try

-  Contains code that may cause an exception.

catch

-  Handles the exception when one occurs inside try.
-  Allows the program to continue after the exception is handled.

Explanation

What happens

-  Valid array indexes are 0, 1, and 2.
-  Accessing `numbers[10]` causes an `IndexOutOfRangeException`.
-  The exception is caught by `catch`.
-  `e.Message` returns the exception message.

Exception `e`

-  Exception → type used to represent the exception.
-  `e` → variable containing information about the exception.
-  `e.Message` → returns a description of the error.

Procedure

- 1 Put potentially failing code inside try.
- 2 If an exception occurs, C# moves to catch.
- 3 Handle the error inside catch.
- 4 Continue with the code after try...catch.

💡 Example

```
try
{
    int[] myNumbers = { 1, 2, 3 };
    Console.WriteLine("Value at index 10: " + myNumbers[10]);
}
catch (Exception e)
{
    Console.WriteLine(e.Message);
}
```

➡ Result

Index was outside the bounds of the array.

Note

- ◆ C# provides specific exception types, such as `IndexOutOfRangeException`, `FormatException`, and `DivideByZeroException`.
- ◆ `catch (Exception e)` handles exceptions broadly.
- ◆ Specific exception types can be used when different errors need different handling.

Comparison

Without try...catch

- ✓ An unhandled exception can terminate the program.

With try...catch

- ✓ The exception is handled.
- ✓ The program can continue after the catch block.

✦ Summary

- ✓ An exception is a runtime error.
- ✓ C# throws an exception when an error occurs.
- ✓ try contains code that may cause an exception.
- ✓ catch handles the exception.
- ✓ Exception e provides information about the error.
- ✓ e.Message gives the error description.

Thank you 😊